Chapter 1
New Advances in Vehicle Routing Problems: A Literature Review to Explore the Future



Mohammad Asghari and S. Mohammad J. Mirzapour Al-e-hashem

Abstract The vehicle routing problem (VRP) is a combinatorial optimization problem that involves finding the optimal design of routes for a fleet of vehicles to serve the demands of a set of customers. This chapter investigates the evolution of three major and applicable streams of the VRPs, namely, electric vehicle routing problem (EVRP), green vehicle routing problem (GVRP), and hybrid vehicle routing problem (HVRP). This survey chapter aims to provide a comprehensive and structured overview of the state of knowledge and discuss important characteristics of the problems, including formulation techniques, methods of solution, and areas of application. We then show how they are different from the traditional vehicle routing problem. Finally, we present a summary table for each variant to emphasize some key features that represent the developments direction of researches.

1.1 Introduction

According to Nakata [144], the future estimation of energy systems indicates that a serious loss will happen by 2040. This future decline in energy availability and the pollution generated by unnatural resources lead to many efforts for preventing the damages by overcoming the pollution resulting from fossil fuel consumption. With technological developments, scientists concentrate on reducing carbon emissions by considering electric or hybrid vehicles as the base of their researches.

Using batteries is the main feature of an Electric Vehicle (EV) which directly influences the performance and cost of the driving [171]. Since the hybrid and electric vehicles are important for reducing environmental damage, researchers have been examining them even though they are still in limited use [82]. As we know, the research efforts before 2010 mainly focused on the traditional VRP. To study more details on the variants of traditional Vehicle Routing Problem (VRP) and

M. Asghari (✉) · S. M. J. Mirzapour Al-e-hashem
Department of Industrial Engineering and Management Systems, Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran
e-mail: mohammad.asghari@aut.ac.ir; mirzapour@aut.ac.ir

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their features, refer to [21, 142]. A few studies on green routing problem especially EV had been conducted during this period. After 2010, routing problems covering energy consumption and pollution emissions started to draw close attention for researchers and became a hot topic in recent years.

We have examined and analyzed the articles in three major and applicable streams, namely, electric vehicle routing problem (EVRP), green vehicle routing problem (GVRP), and hybrid vehicle routing problem (HVRP). As shown in Table 1.1, about 236 papers are reviewed in this chapter that reflects the environmental sensitivity consideration in routing problems and one can observe that this concern is increasing day by day.

The chapter is organized as follows. Section 1.2 is devoted to a summary of recent publications in EVRP, while Sect. 1.3 contains the review on GVRP. This is

Table 1.1 The scientific contribution reviewed in this chapter

Year	Number of papers	The list of the papers
1976	1	[35]
1998	1	[12]
1999	1	[50]
2000	2	[144, 165]
2002	1	[161]
2004	1	[176]
2005	2	[13, 115]
2006	1	[67]
2007	6	[33, 117, 137, 149, 179, 186]
2008	6	[8, 72, 74, 135, 145, 198]
2009	3	[20, 190, 203]
2010	8	[6, 15, 51, 58, 88, 126, 131, 187]
2011	8	[17, 39, 52, 73, 160, 194, 202, 217]
2012	8	[41, 48, 83, 85, 102, 175, 213, 215]
2013	9	[10, 54, 60, 61, 127, 143, 150, 205, 208]
2014	20	[2, 3, 38, 42, 43, 53, 63, 77, 87, 89, 95, 98, 112, 148, 151, 153, 164, 177, 184, 227]
2015	21	[5, 9, 24, 25, 36, 45, 68, 70, 71, 100, 109, 111, 122, 123, 142, 185, 192, 200, 214, 218]
2016	34	[4, 16, 18, 21, 26, 31, 34, 40, 44, 59, 69, 76, 78, 79, 82, 90, 94, 103, 108, 113, 124, 140, 146, 154, 158, 166, 167, 171, 174, 193, 195, 199, 206, 212, 221, 222]
2017	39	[1, 7, 11, 19, 27, 28, 32, 37, 49, 57, 62, 64, 65, 81, 84, 101, 110, 120, 132, 134, 139, 141, 155, 156, 159, 168, 173, 178, 180, 181, 188, 191, 197, 201, 219, 220, 226, 228, 235]
2018	34	[22, 29, 46, 55, 91, 91, 96, 97, 99, 104, 105, 114, 116, 119, 130, 133, 136, 138, 147, 152, 169, 182, 183, 189, 196, 207, 209, 229–231, 236]
2019	29	[14, 23, 30, 47, 66, 75, 80, 86, 93, 106, 107, 118, 121, 125, 128, 129, 157, 163, 170, 172, 204, 210, 211, 216, 224, 225, 232–234]

followed by the surveys on HVRP in Sect. 1.4. To make this easier for interested readers, a table of summary on mode of formulation, methods of solution, and other essential features is given in each review section. Eventually, Sect. 1.5 provides the main findings of the literature review and directions for further research.

1.2 Electric Vehicle Routing Problem

In the last decade, EVs have been proposed as alternative-energy transportation for personal and commercial uses, and their market share has grown rapidly [54]. The environmentally friendly aspects of EVs, such as low noise pollution, no greenhouse gas emissions, and high-energy efficiency is the main reason of increase in their popularity that benefits logistics firms in achieving a green image from a growing number of environmentally and socially knowledgeable clients [44]. Moreover, state regulatory support is an influential factor that encouraged using EVs in both individual travel and public transportations [103]. Figure 1.1 displays the obvious increase experienced during the last years in the number of EV-related articles published in Scopus-indexed journals, which proves the growing interest that the use of EVs in routing problems is arising among researchers and practitioners. Readers can find recent sustainability investigation and realistic evaluation techniques of life cycle carbon emission generated by electric vehicles in [31, 55]. Along with the environmental advantages, these kinds of vehicles are considered attractive than conventional fossil fuel-powered vehicles in terms of a cost perspective. In

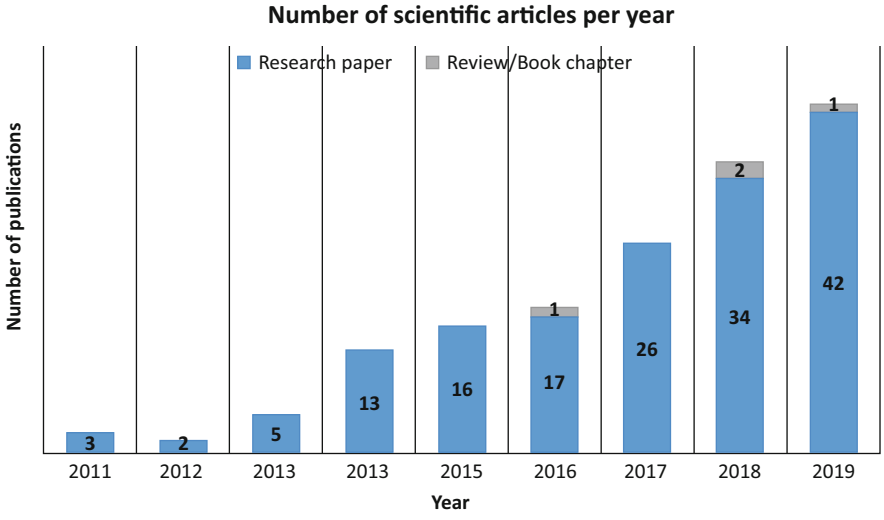


Fig. 1.1 Evolution of EVRP related publications in Scopus-indexed journals

comparison to the cost of petrol/gasoline consumed by a conventional vehicle, when traversing the same distance, an EV costs 85–90% less.

Despite comprehensive researches on the characteristics and attributes of EVs and the design of charging infrastructure, the network modeling of EVs is still developing and limited. Because of the assuring opportunity of the EVs for reducing pollution impacts and transportation costs compared with fossil fuel based engines, the EVRP, which deals with planning routes for commercial EVs, is one of the logistics distribution/service field investigated in the recent years by researchers [29, 80, 113, 153].

Table 1.2 classifies the main features of the studies done on EVRP as an extension to the review provided by Keskin et al. [93]. The headings of the column represent recharging function (Rch), the function of electricity consumption (EC), fleet type, charger composition in the stations, charge amount, considering time windows (TW), determining the location of station (Lc), objective function (Obj.), and solution approach; linear and nonlinear recharging time functions are indicated by L and N; homogeneous and heterogeneous fleets are indicated by Ho and Ht; NS, *, **, ***, **** stand for not specified properties, consumption depends on speed, mass and gradient of the terrain, linear charging with different cost components for different charge levels, linear charging with different cost and rates for different chargers, linear and constant charging for battery swapping and parallel with considering service time, respectively; A-, G-, AC, CG, DP, E-VNS, HC, ILS, PBA, PSO, TS, and WSM correspond to adaptive-, general-, ant colony, column generation, dynamic programming, evolutionary variable neighborhood search, heuristic concentration, iterated local search, path-based approach, particle swarm optimization, tabu search, and weighted sum method, respectively. CS is also used for the articles that just apply a commercial solver for reporting results. Table 1.3 defines the encoding used for each objective function.

The main difficulty associated with planning the EVs is how to include the visiting stations into the optimization, because the EVs have short travel range and their recharging can be time-consuming and uncomfortable during the travel. Some of the studies incorporate recharging at the stations during the travel, such as in the green routing problems [7, 48, 53], partial charging EVRP [28, 90, 182], battery swapping EVRP [81, 86, 218], EVRP with a mix of partial charging and battery swapping [86, 152, 181], and the EVRP considering a nonlinear recharging time function [64–66, 141]. Macrina et al. [128] present a table providing most of the recent mathematical models that use linear distance-dependent energy consumption.

Chargers in the stations may be equipped with different options including power voltage and maximum current, which have an effect on the duration of charging. Some models take stations with different chargers into account [53, 91, 111, 177]. Additionally, some of researches considered the heterogeneous fleet composed of various EVs with different features [44] whereas other researches cover different internal combustion engine vehicles (ICEVs) as well [71, 80, 97, 177]. Recently several studies have incorporated the location of the stations in the routing optimization [111, 152, 181, 218].

Table 1.2 EVRP literature overview [93]

	Rch		EC		Fleet type		Charger composition		Charging					
Papers	L	N	L	N	Ho	Ht	Single	Multi.	Full	Partial	TW	Lc	Obj.	Solution method
[39]			✓		EVs		✓		✓	✓			1, 2, 8, 15	Iterative construction, improvement heuristics
[48]			✓		AFVs		✓		✓				2	Several heuristics
[53]	✓		✓		EVs			✓		✓	✓		2, 15	Several heuristics and SA
[177]	✓		✓			EVs, ICEVs		✓		✓	✓		1, 2, 15	Several heuristics
[184]	✓		✓		EVs		✓		✓		✓		1, 2	Hybrid VNS and TS
[9]	✓		✓		PHEVs		✓			✓			6, 9, 10	DP
[25]	✓		✓		EVs		✓			✓	✓		1, 2, 4, 15	VNS branching
[45]	✓		✓		EVs		✓			✓	✓		2	Hybrid VNS and TS
[71]	✓		*			EVs, ICEVs	✓			✓	✓		2, 8, 9, 10, 15	ALNS
[111]	***		✓		EVs			✓	✓		✓	✓	5, 8, 15	Hybrid AVNS and TS
[218]			✓		EVs		BSS					✓	2, 5	Several heuristics
[44]	✓		✓		EVs		✓		✓	✓	✓		2	Branch-price-and-cut
[76]	✓		✓		EVs		✓			✓	✓		1, 2, 15	WSM
[79]	✓		✓			EVs	✓		✓		✓		1, 2	Branch-price, and ALNS
[90]	✓		✓		EVs		✓			✓	✓		1, 2	ALNS
[94]			✓		AFVs		✓		✓				2	Branch-and-cut
[108]			NS		EVs		BSS						8	DP
[113]	✓		✓		EVs		✓		✓				1, 2, 15	CS
[140]			✓		AFVs		✓		✓				2	MSH
[174]	✓		✓		EVs		✓		✓	✓	✓		2	GVNS, DP
[212]	✓		✓		EVs		✓			✓	✓		1, 2	ALNS
[7]			✓		AFVs		✓		✓				2	An exact algorithm
[27]	✓		✓		EVs		✓			✓	✓		1, 8	VNSB based matheuristic
[64]		✓	✓		EVs		✓			✓			8	Two-stage matheuristic
[81]			✓		EVs		BSS					✓	2, 5	AVNS

(continued)

Table 1.1 (continued)

Papers	Rch		EC		Fleet type		Charger composition		Charging		TW	Lc	Obj.	Solution method
	L	N	L	N	Ho	Ht	Single	Multi.	Full	Partial				
[101]	✓		✓		AFVs		✓			✓			2	CS
[141]		✓	✓		EVs			✓		✓			8	Hybrid ILS and HC
[181]	✓		✓		EVs		✓		✓	✓	✓	✓	1, 2, 5	CS
[197]	**		✓		EVs			✓		✓			2, 4, 15	Several exact and heuristic methods
[22]	✓		✓		EVs		✓		✓				1, 2	LNS, Exact algorithm
[29]			✓		AFVs		✓		✓				2	An exact method
[91]	✓		✓		EVs			✓		✓	✓		1, 15	ALNS based matheuristic
[96]		✓	✓		EVs			✓		✓		✓	5, 8	ALNS based matheuristic
[97]	✓			✓		EVs, ICEVs	✓			✓	✓		11	CS
[99]		✓	✓		EVs		✓			✓			7	Several exact and heuristic methods
[104]	✓		✓		EVs		Single chargers & WCS			✓			8, 15	CS
[130]			✓		AFVs		✓		✓	✓	✓		2	CS
[136]			*			EVs	BSS				✓		2	E-VNS
[152]	****		✓		EVs		BSS		✓	✓	✓	✓	1, 2, 5	CS
[182]	✓		✓		EVs		✓			✓	✓	✓	1, 5, 8	ALNS, DP
[207]		✓	✓		EVs, ICEVs			✓		✓	✓		1, 2, 6, 15	GRASP based matheuristic
[209]			✓		EVs		BSS		✓		✓		12	Heuristic procedures
[230]	✓		✓		EVs		✓			✓	✓		1, 2, 4, 8, 15	TS
[14]	✓		*		EVs		✓		✓		✓		3	Bellman–Ford algorithm
[30]			✓		AFVs		✓		✓				2	PBA
[66]		✓	✓		EVs		✓			✓			8	CS
[80]	✓		✓			EVs, ICEVs, PHEVs	✓			✓	✓		1, 9, 15	GA, LNS with IP solver
[86]			✓			EVs	BSS			✓			2, 13, 14	Hybrid CG and ALNS
[128, 129]	✓		✓		EVs, ICEVs		✓			✓	✓		2, 15	ILS
[157]			✓		EVs								16	Two-phase based LNS

Table 1.3 Explanation of the objective functions used in Tables 1.2 and 1.5

1: Vehicle cost/wage	9: Battery cost
2: Total distance/travel cost	10: Charging cost
3: Fuel consumption/emission	11: Profit/revenue
4: Waiting cost	12: Operational costs
5: Station installation cost	13: Battery swapping cost
6: Stopping cost at a station	14: Unit time penalty for violated time windows
7: Total makespan (duration)	15: Recharging cost
8: Fuel cost	16: Fixed and maintenance cost

In the majority of papers, recharging is done at special charging stations, although several studies investigate replacing discharged battery with a full charged one which is called battery-exchange stations [86, 108, 136, 152, 209], and recharging the EV by an inductive charging system located along the roads which is known as wireless charging systems [104]. Recently, some papers have studied the context of the two-echelon EVRP [22, 86], reverse logistics [231], technician routing [207], and pickup and delivery problem [76, 130]. A comprehensive review of the goods distribution with EVs can be found in [154, 155].

Conrad and Figliozzi [39] presented the first research on the EVRP where EVs can be recharged at some client places to continue their trip. The model had a fixed charging time by allowing fully charge or maximize 80% of the energy capacity at just client places. Since then, several extensions have been proposed. In the EVRP, aside from battery capacity, other factors such as considering time windows for customers and load capacities for vehicles are repeatedly taken into account. Conrad and Figliozzi suggested a bi-objective mathematical formulation that considers EVs. The objectives aim to minimize the total traveling distance and the number of routes. To solve the proposed problem, the authors offered an iterative route construction and improvement algorithm. Worley et al. [213] investigated a mixed-integer mathematical formulation for the EVRP with charging station siting constraints to concurrently minimize the total cost of traveling, recharging, and station location. Wang and Cheu [208] studied the recharging operations of a fleet of electric taxies. Then, to minimize the total distance traveled and maximum route time, they proposed a TS algorithm.

Currently, a few related papers in the context of EVs have addressed the location-routing problem in an uncertain environment. For example, a robust optimization structure was applied by Schiffer and Walther [182]. Their study analyzes uncertain parameters consisting of demand, customer locations, and time windows in a simultaneous problem of locating charging stations and routing EVs. In this context, Lu et al. [114] suggested a multi-objective mathematical model that considers load dispatch of a microgrid accessing stochastically to a large-sized of uncoordinated charging of EVs. The objective functions aim to minimize the operating cost, pollutant treatment cost, and load variance. The authors proposed an enhanced PSO algorithm to solve the problem effectively. Under three different

scheduling scenarios, the dispatch results showed shifting charging loads from high-priced periods to low-priced periods in the presence of the coordinated charging mode of EVs.

Regarding the uncertain service time spent at the stations, Sweda et al. [197] investigated the stations under the probability unavailability. When an EV reaches an occupied station, it may need to wait for a random time to get service. A relevant problem was solved by Froger et al. [64] who determined an inadequate number of chargers for each station and a vehicle has to wait for recharging while other EVs in the fleet are using the chargers. Although in the proposed problem, the routing and charging features affect the use of the chargers, the queue lengths do not depend on these decisions. The waiting times at the alternative-fuel stations (AFSs) were recently studied by Bruglieri et al. [29]. The authors planned the routing of the alternative-fuel powered vehicles (AFVs) so that the consumed times for refueling of the AFVs do not overlap in the fuel stations. Finally, an exact approach is used to minimize the total distance traveled. Kullman et al. [99] considered a public–private recharging policy for the EVRP in which a charger in the depot belongs to the company, while the vehicles can also be recharged at public stations. If the EVs recharge at public stations, they may queue using an $M/M/\psi_c/\infty$ queuing system, otherwise use the depot charger as soon as they arrive. The results illustrated a substantial decrease in the cost of public–private strategies compared to private-only strategies.

Considering the literature, many different strategies exist for solving the EVRP and its variations. In a strong sense, the EVRP can equally be recognized as NP-hard problem because the EVRP is a generalization of the VRP, which is known as an NP-hard [44, 230]. Hence, metaheuristic optimization algorithms, which are earlier introduced for global optimization problems, are applied in order to solve most of the studies in this field. Table 1.4 reviews the metaheuristic algorithms-based solution approaches employed in the EVRP context. To improve algorithm efficiency, the integrated approaches, such as column generation, metaheuristic, dynamic programming, etc., have been applied as a subroutine in the algorithms. It is worth noting that most of the papers considered large neighborhood search and population-based algorithms like genetic algorithm, although a permutation order-based coding system can allow rendering a solution without doing any transformation.

Schneider et al. [184] done the first attempts for considering the EVRP with time windows, referred to as EVRPTW for simplicity, in which time consumed for recharging depends on the amount of energy left upon arrival at a recharging vertex. Recharging depends on the battery state of charge (SOC). It continues until the battery capacity is fully charged, which its duration is assumed to be commensurate with the amount of the energy transferred. The authors used a hierarchical objective, which minimizes the number of vehicles first and the total traveled distance second. To solve the proposed EVRPTW, a hybrid metaheuristic VNS/TS algorithm is developed and its performance is examined by several benchmark data sets related to VRPs. Goeke and Schneider [71] extended the same problem which incorporates a mixed fleet of EVs and ICEVs. The objective function aims to minimize the

Table 1.4 Metaheuristic optimization algorithms employed in the EVRP

Study	ACO	CS	DEA	GA	ILS	A-/LNS	MA	SA	A-/TS	A-/E-/VNS	Combined with
[208]									✓		
[53]								✓			
[148]				✓							TS
[164]									✓		
[184]										✓	TS
[5]				✓							TS
[24, 25]										✓	Matheuristic
[45]										✓	TS
[71]						✓					
[111]										✓	TS
[218]				✓							Expert knowledge
[78]						✓					Dynamic programming
[79]						✓					Dynamic programming
[90]						✓					
[158]					✓						Set partitioning
[174]										✓	Dynamic programming
[212]						✓					
[1]				✓							Markov decision process
[11]			✓								
[28]										✓	Matheuristic
[81]										✓	
[141]					✓						Heuristic concentration
[173]							✓				GA
[188]				✓							Dynamic Dijkstra
[235]				✓							
[22]						✓					
[116]						✓					
[91]						✓					
[92]						✓					Matheuristic
[96]						✓					Matheuristic
[136]										✓	
[183]						✓					Dynamic programming

(continued)

Table 1.4 (continued)

Study	ACO	CS	DEA	GA	ILS	A-/LNS	MA	SA	A-/TS	A-/E-/VNS	Combined with
[182]						✓					Lower bounding procedure
[189]				✓							Dynamic Dijkstra
[230]	✓										ILS
[231]									✓		
[236]		✓									
[23]						✓					
[80]				✓							LNS
[86]						✓					Column generation
[128, 129]					✓						
[125]								✓			TS

ACO ant colony optimization, *CS* cuckoo search, *DEA* differential evolution algorithm, *GA* genetic algorithm, *ILS* iterated local search, *A-/LNS* adaptive-/large neighborhood search, *MA* memetic algorithm, *SA* simulated annealing, *A-/TS* adaptive-/Tabu search algorithm, *A-/E-/VNS* adaptive-/evolutionary-/variable neighborhood search

total cost determined as a function of speed, gradient, and cargo load. Additionally, Desaulniers et al. [44] tackled EVRPTW by studying various charging policies and applied branch-price-and-cut algorithms to solve them optimally. Hiermann et al. [79] extend a mathematical model of the EVRPTW presented by Schneider et al. [184] by analyzing a heterogeneous fleet of different EVs. They minimize the total distance traveled as well as acquisition costs for each vehicle that leaves the depot. Both studies used ALNS as the solution methodology. Further solution techniques for the EVRPTW concentrating on a math heuristic and a VNS branching are introduced by Bruglieri et al. [24, 25].

Earlier EVRPTW studies by Afroditi et al. [3], Schneider et al. [184], and Bruglieri et al. [24, 25] considered charging at some specific points on the route but neglected recharging at client locations as well as simultaneous partial recharging and siting decisions. Later, the EVRPTW was extended by several authors with different assumptions and formulations. In [164], the authors addressed a load-dependent function for energy consumption in delivery service systems, which was solved by an ATS algorithm. Instead of the battery charging operations of the EVs, Chen et al. [34] studied operations for battery swapping operations. Grandinetti et al. [76] investigated a multi-objective mixed-integer linear programming (MILP) mathematical model for a pickup and delivery problem with soft time windows, through a fleet of EVs. Such a problem tries to minimize the total traveled distance, the vehicle cost, and the penalty cost for violated time windows. Paz et al. [152] considered a multi-depot EVRPTW. Recently, Xiao et al. [216] modeled the EVRPTW considering the maximum range of an EV or the energy/electricity consumption rate (ECR) per unit of distance traveled by an EV dynamically

according to speed and load along the route. Keskin et al. [93] investigate time-dependent queueing times at the stations for the EVRPTW with allowing but penalizing late arrivals at the depot and client sites.

Roberti and Wen [174] is the first attempt concentrating on the electric traveling salesman problem with time windows (ETSPTW) which is a special case of the VRP with only a single EV. Since the ETSPTW is known as a relatively new research field, there are just a few studies on the TSP optimization of EVs in the literature. Küçükoglu et al. [125] considered both private and public recharging stations for the ETSPTW and extend it by additionally considering recharging operations at client sites with various recharging rates, called hereafter the ETSPTW with different recharging rates.

1.2.1 Refueling Policy

There are three recharging ideas about departure state-of-energy in the tank and the time spent for at a fuel station. In "Fixed-full" refueling approach, which fits BSSs, it has been assumed that a vehicle leaves a fuel station when its tank capacity is filled in a particular constant duration. "Variable-full" approach has a same assumption regarding departure with full state-of-energy but relaxes the constant time assumption. Time spent on refueling under this approach is proportional and depends on the energy required for filling the tank capacity. "Variable-partial" approach allows a vehicle to depart with partial state-of-energy and has a proportional fueling time to energy gain. Below we summarize the main studies on EVs charging policies.

1.2.1.1 Full Charging Policy

A full recharging strategy means that an EV is allowed to leave a charging station contingent upon its battery capacity is fully charged. Erdogan and Miller-Hooks denoted the problem as GVRP with a Fixed-full approach, in which the tank capacity of AFV is filled at AFSs within a constant amount of time. Intending to minimize the total distance, the model seeks to eliminate the risk of running out of fuel. They consider the service time of each customer and the maximum duration restriction was posed on each route. Schiffer and Walther [181] analyzed both full and partial charging options for EVs in their location-routing model of charging stations, as well as time window constraints. The objective function aims to minimize not only the traveled distance but also the number of charging stations and the number of used EVs. The researches of Conrad and Figliozzi [39], Schneider et al. [184], Li-Ying and Yuan-Bin [111], Desaulniers et al. [44], Hiermann et al. [79], Koç and Karaoglan [94], Lin et al. [113], Montoya et al. [140], Roberti and Wen [174], Andelmin and Bartolini [7], Breunig et al. [22], Bruglieri et al. [29, 30],

Madankumar and Rajendran [130], Paz et al. [152], Wang et al. [209], Zhang et al. [229], and Basso et al. [14] can be mentioned here.

1.2.1.2 Partial Charging Policy

Aside from the full charging assumption, some researches follow partial charging policies where an EV may leave a recharging station with full battery capacity or any level depending on the time spent for recharging. The full charging policy was first relaxed and the EVRP was modeled by allowing EVs to recharge partially by Felipe et al. [53] who applied various recharging technologies for the EVs. The proposed problem aims to minimize the overall costs consisting of costs for the total distance and recharging costs. Various cost terms are supposed for different recharging decisions as the authors assumed that recharging overnight at the depot is more economical than recharging along the route. Besides, many costs and recharging options are surveyed for the different charging technologies including slow and fast charging. Nevertheless, the authors do not take charging at client locations into account and neglect simultaneous siting decisions. To solve the problem, they developed three heuristics including a deterministic local search technique, a construction heuristic, and simulated annealing algorithm. As reported by their computational results, the SA operates more reliably concerning other algorithms for large-scale instances. Moreover, the impact of fast charging options in the presence of time windows has also been investigated by Çatay and Keskin [32].

Further research on partial charging is tackled by Desaulniers et al. [44] concentrating on four different recharging policies per route including only a single full charge, only a single partial charge, multiple full charges, and multiple partial charges. To solve the EVRPTW, they performing an exact method, branch-cut-and-price algorithm. Likewise, Keskin and Çatay [90] investigated the same problem with a partial recharge strategy and developed an ALNS algorithm to solve the problem. The computational results show the better outcomes of the full recharging strategy as well as the benefits of the partial recharging strategy which are pointed out by comparing the partial and full recharging policies. A similar strategy can also be found in [24, 25, 79] that relaxed the full recharge restriction and allowed batteries to be recharged up to any level along the travel route. The former examined also a partial charging strategy in the EVRPTW, where the objective functions minimize the number of used EVs and the time of traveling, waiting, and recharging. To solve small size instances, they developed a VNS branching algorithm. Later, in [25], they model EVRP considering different EVs with various capacities, costs, and recharge rates.

Bruglieri et al. [28] is another research regarding partial recharging strategy where a three-phase metaheuristic approach combined with an exact method and a VNS local branching is presented to solve the proposed EVRP. A different assumption for the EVRP is taken into account by Schiffer and Walther [180, 181] and Schiffer et al. [183] in which siting decisions for the recharge stations are also investigated simultaneously. The first study that extended EVRP to consider

nonlinear charging functions is presented by Montoya et al. [141]. The objective function aims to minimize the total time of the operations consisting of charging times and travel times. To solve the problem, the authors developed a hybrid metaheuristic and introduced new benchmark instances. New formulations of this problem are considered by Froger et al. [64, 65]. They developed two new extensions of the EVRP. In the first study, the authors considered nonlinear charging functions, while in the second one, they dealt with an EVRP with capacitated recharging stations.

1.2.1.3 Battery Swapping

In the literature, some researchers assumed that the EVs have swappable batteries that can exchange immediately in battery swap stations (BSS), thereby avoiding a long recharging time. Adler and Mirchandani [2] considered a case study of EVs application in an online routing system, where the authors try to optimally route the multiple origin-destination pairs while studying battery swapping and reservation. Zheng et al. [227] introduced an optimal design of battery swap/charging stations for a distribution system. Yang and Sun studied capacitated EVs in a location-routing problem with BSSs, which concentrates on simultaneously determining the routing plan of EVs and the location of BSSs with considering limitations for battery driving range [218]. The EVs always depart from stations with a full battery with considering BSSs instead of charging stations. Besides, in the basic formulation, a BSS can be visited at most once, whereas their proposed problem allowed a BSS to be visited several times.

Another EVRP with battery swapping operations was studied by Liao et al. [108], who tried to find a tour involving the origin and all given points while visiting a set of expected BSSs. To solve the extended model, they presented polynomial-time algorithms that consider the problem with fixed tours where all points are visited in fixed sequences. The EVRP model suggested by Yang and Sun [218] was developed by Schiffer and Walther [181] to consider different effective factors including time window, partial charging policy, and stopping at charging station. Subsequently, Hof et al. [81] investigated a BSS location-routing problem for EVs with intermediate stops and developed an adaptive VNS approach to solving it. Also, Paz et al. [152] modeled the same problem considering multi-depots and battery swapping option in some locations.

Recently, Zhang et al. [232] improved previous deterministic routing models by developing a stochastic routing model applicable to the actual stochastic occurrences. They present an electric vehicle BSS location-routing problem with stochastic demands, intending to determine a minimum cost scheme including the optimal number and location of BSSs with an optimal route plan based on stochastic customer demands. Furthermore, the classical recourse policy and preventive restocking policy are extended by considering the influences of both battery and vehicle capacity simultaneously. To solve this problem, a hybrid approach was proposed that combines the binary PSO algorithm and VNS algorithm.

1.2.2 Energy Consumption Rate

Because of the limited battery capacity and the resulting shorter achievable range of EVs, such routing problems require new approaches that can accurately monitor the vehicles' energy consumption to ensure they never get stranded. For example, Barco et al. [10] proposed an EVRP, which minimizes the used energy instead of the traveled distance. An approximated energy consumption model based on physical dependencies is used to compute energy consumption. Since the model is applied to an airport shuttle service that carries passengers from the airport to a hotel and back at known times, recharging is only possible at the depot. They conduct a scheduling model for this application. In this model, the recharging of vehicles on known routes is scheduled, taking battery degradation effects into account. Due to the application, the approach does not regard the usage of charging stations on routes or any siting decisions.

Some studies modeled the EVRP by a linear battery discharges rate concerning traveled distance. For example, Felipe et al. [53] assumed various technologies for battery charging where each of them is related to a constant charging rate and has a fixed unit cost. Others have considered a more detailed energy consumption model based on the comprehensive emissions model of Barth et al. [13] which computes the required energy to traverse any given arc based on certain vehicle and arc characteristics (e.g., mass, speed, elevation, acceleration, rolling friction, frontal area, air drag). This energy consumption model was first used in a VRP setting for the pollution-routing problem to estimate fuel consumption and emissions [17], and subsequently in many EVRP studies to track the state-of-energy consumption [14, 71, 100, 164].

In all of the discussed studies, there is a simple assumption about the energy consumed by a vehicle over a route which is assumed to be a linear function of the traveled distance. The earliest research on the modeling of a more realistic energy consumption approach for the EVRP was presented by Goeke and Schneider [71] who designed a nonlinear ECR for a mixed fleet of EVs and fuel vehicles. Although they considered practical factors such as the gradient of the terrain, the vehicle speed and load, the travel speeds/times are considered constants and previously known. Therefore, only the weight of the cargo affects energy consumption over a route. To determine the energy consumption of EVs, Fetene et al. [57] harnessed big data and showed that the ECR may change considerably with a nonlinear behavior depending on the driving patterns and weather conditions.

Lin et al. [113] is another study that developed energy consumption model for EVs, which considered the effects of vehicle load on battery consumption. Fiori et al. [59] introduced a comprehensive power-based electricity consumption formulation that can estimate the instantaneous ECRs by evaluating inputs of EVs (vehicle mass as well as vehicle speed and the instantaneous acceleration). A physical-based analytical approach for calculating energy consumption was developed by Yuan et al. [226]. The authors traced ECRs based on the energy flow path of EV during a driving cycle which includes driveline losses, electric motor

losses, battery losses, load losses, and brake losses. Liu et al. [110] studied the effect of the road gradient on the energy consumption of an EV. Pelletier et al. [156] integrated battery degradation into the model and optimized charging schedules at the depot. They also provided managerial insights considering degradation, grid restrictions, charging costs, and charging schedules of the fleet.

To the best of our knowledge, only Fontana [60] has considered the uncertainty surrounding the energy consumption of electric vehicles and has modeled the problem within a robust optimization framework. However, this author considers the idea of uncertainty surrounding the energy consumption of EVs in an optimal path problem setting, while Pelletier et al. [157] extended it to an EVRP setting in which the vehicle loads influence energy consumption.

1.2.3 Fleet Size and Mix

As EVs have gained popularity worldwide over the past decade, the issues related to the usage of EVs for transportation have opened up a wide range of relevant research works. In addition to the mentioned strategy for EVs, some of the researchers pay attention to the impacts of fleet characters on routing plans and incurred travel costs. Contrary to Van Duin et al. [205] which did not consider battery charging, assumed charging could occur anywhere [73] or only at the depot between routes [100], other fleet size and mix studies incorporating EVs have considered the possibility for the vehicles to stop at charging stations during their routes and have taken into account the actual location of the stations.

Pelletier et al. [154] provided researchers with the technological and marketing background of goods distribution with a fleet of EVs and presented a review of existing research works in the EVRP field. Their overview summarized the main existing research works on EVs in transportation science, which covers three parts: fleet size and mix, vehicle routing, and optimal paths. From their work, they found that EVs have many advantages such as frequent stop-and-go movements, low travel speeds and reduced pollution, and noise in last-mile deliveries of urban logistics.

A different variant of the problem with a mixed fleet of EVs and conventional ICEVs was developed by Goeke and Schneider [71] who used realistic energy consumption functions for both types of vehicles, considering vehicle mass as well as vehicle speed and terrain gradients. Three different objective functions are presented. The first objective minimizes the total traveled distance, whereas the second objective minimizes overall costs as the sum of energy costs and wages for drivers. The third objective extends the second one by adding costs for battery lifetime. However, options of partial recharging and simultaneous siting decisions are not taken into consideration.

Lebeau et al. [100] have studied the fleet size and mix EVRPTW. The fleet is composed of EVs and ICEVs each having their fixed costs, operating costs, payload, volume capacity, energy capacity, and energy consumption. The objective is to minimize the sum of all fixed, operating, and driver costs. Each customer has

a demand, a time window, and a service time. The depot also has a service time representing loading time so that vehicles can accomplish several routes. The EVs can be charged at the depot during loading with a linear charging rate. They solved the problem using a modified Clarke and Wright Savings (MCWS) algorithm.

Another study that considers different types of EVs in a heterogeneous fleet for customer visits was done by Küçükoglu and Öztürk [124]. Compared with the computational results on small-scale sample sets generated via the mathematical formulation suggested by Schneider et al. [184], the authors pointed out the benefits of using the heterogeneous fleet on the number of used EVs and the total traveled distance. To solve the same problem with a heterogeneous fleet, Penna et al. [158] proposed a hybrid iterative local search (ILS) algorithm developed by combining an ILS algorithm with a set partitioning problem, and Hiermann et al. [79] offered an efficient ALNS algorithm whose performance has been demonstrated in several sample sets.

1.3 Green Vehicle Routing Problem

As the energy overuse and generated pollution have a potential threat for our environmental and ecological condition, many researchers have taken the initiative way to join the green campaign to prevent more damage to the environment. Although barriers to achieving sustainable solutions still exist, energy policies need to be changed to reduce fossil fuel consumption and encourage the use of AFVs. In the last decade, regarding the growing knowledge about the hazardous effects of logistics activities, such as transportation which has a considerable impact on the ecological environment, requests for more environmentally friendly practices have increased [112]. In this way, Erdogan and Miller-Hooks [48] introduced the GVRP, a relatively new research field, as an extension of the traditional routing problem. To decrease energy consumption in fleet operations, GVRPs aim to minimize fuel consumption or adopt AFVs solely or with existing conventional gasoline or diesel vehicles (GDVs) in fleets.

This section illustrates transportation and energy consumption, followed by a comprehensive survey on the recent investigations on the pollution-routing problem. For earlier surveys, interested readers can refer to [112]. Figure 1.2 displays the trend of scientific contributions (research papers, review and book chapters) on GVRP published in Scopus-indexed journals. As shown in the figure, over the years, the green concept and its several variants have received increasing attention in the scientific context, especially in the field of operations research [18, 43, 112, 151].

Table 1.5 reviews the relevant works in terms of the characteristics of the mathematical model, fleet type, fueling approach, and solution method. Definitions of the numbers associated with each objective function have been presented in Table 1.3.

As shown in Table 1.5, most GVRPs examined vehicle weight, speed, and travel time as effective factors on greenhouse emissions [18, 112, 151, 225]. Various

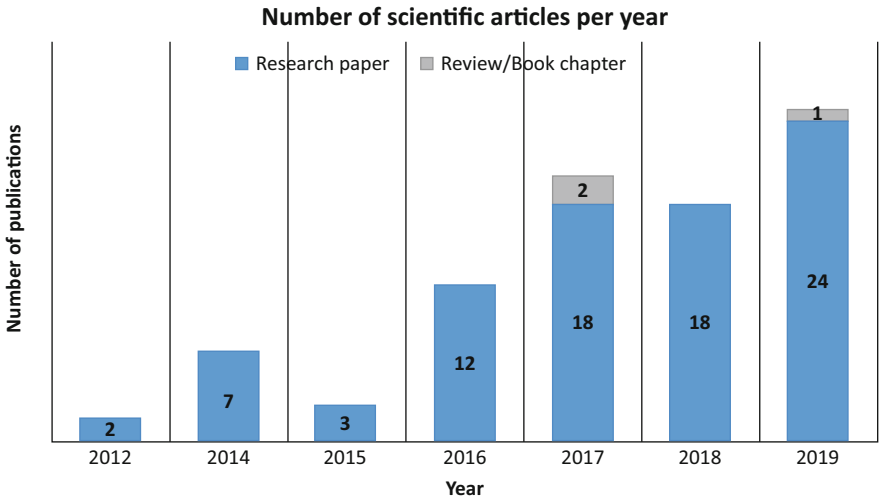


Fig. 1.2 Studies in the context of GVRP and its variants

VRP characteristics have been incorporated into the GVRP literature. These include customer time windows and vehicle capacities [27, 71, 79, 90, 184, 185], multiple charging technologies [53], speed- and/or load-dependent energy consumption [71, 162], nonlinear recharging times [141], recharging station congestion [45, 64, 65], time- and/or station-dependent refueling costs [56, 91].

Bektas et al. [18] studied GVRP where CO_2 emissions are explicitly taken into account. They reviewed some recent developments in the field, including the description of some vehicle-emission models and their applications in road freight transportation. To minimize the carbon emissions for a green pickup and delivery problem with the lowest carbon emissions, Yu et al. [222] applied integrated scheduling. Qazvini et al. [166] proposed a multi-depot green pickup and delivery problem along with time windows to lessen fuel consumption while a GVRP with cross-docking was investigated in [221]. Sawik et al. [178] concentrated on multi-objective GVRP optimization consisting of the minimization of pollution, noise, and fuel consumption. Wang et al. [209] addressed the compensation and profit distribution by using the cooperative game theory for the cooperative green pickup and delivery problem with the lowest carbon emissions.

In comparison with other variants of GVRPs, the literature that considers halts for refueling/recharging at fuel/charging stations is limited. Erdogan and Miller-Hooks [48] examined the possibility of recharging or refueling a vehicle with environment-friendly fuel like biogas as well as the halts for refueling at stations. They conceived the main idea from Bard et al. [12] who addressed VRP with considering maximum carrying load capacity for the fleet.

Lin et al. [112] considered charging stations in the tour to overcome the capacity limitation of the fuel tank. The approach of Chung and Kwon [36] considers the

Table 1.5 Summary of the previous work on the GVRP

Study	Fleet type		Capacity	Load	Speed	Time window	Multi-obj	Objective(s)	Solution method
	Ho	Ht							
[63]	AFV		✓	✓		✓		3	An exact algorithm
[122, 123]	GDV		✓	✓	✓	✓	✓	1, 3	ILS based metaheuristic
[185]	AFV		✓			✓		2	AVNS
[69]	GDV		✓	✓	✓	✓		1, 3	Exact method
[195]	GDV		✓	✓	✓		✓	3	SA & TS
[37]	GDV		✓	✓	✓		✓	3, 8	Dynamic programming
[49]	GDV		✓	✓	✓	✓		3	Exact method
[62]	GDV		✓	✓	✓	✓	✓	1, 3	ALNS
[84]	GDV		✓	✓			✓	2, 3	ACO & VNS
[191]	GDV		✓	✓	✓			2	Dynamic programming
[201]	GDV		✓	✓	✓		✓	3, 12	Exact method
[219]		AFV & GDV						2	Heuristic procedure
[220]		AFV & GDV	✓					2, 3, 8	VNS
[119]	GDV		✓	✓	✓	✓	✓	2, 3, 12	NRGA, NSGA-Π & PSO
[138]		GDV	✓	✓	✓			2	Exact method
[147]	GDV		✓	✓	✓	✓		1, 3	TS
[162]	GDV				✓		✓	2, 3	PSO
[229]	GDV		✓	✓	✓	✓		3, 8, 11	ALNS
[47]	GDV		✓	✓	✓	✓		3, 12	Local search
[106]	AFV		✓				✓	2, 3, 7, 11	ACO
[121]		AFV & GDV	✓			✓		2	Branch-and-cut
[170]		GDV		✓	✓	✓		1, 3, 7	Exact method
[172]	GDV			✓	✓			2, 3	NSGA-Π
[210]		AFV & GDV		✓				3	TS
[233]	AFV							2	ACO

fact that the installation of a network of stations is a long-term process that usually requires several periods to be completed. Hence, their model restricts the number of newly established stations per period while pursuing maximization of possible traffic over all periods. Bruglieri et al. [26] contributed a new formulation of GVRP which includes a precomputation of AFS to locate between each pair of vertices. Their model avoids using multiple copies of AFS that leads to a decrease in the number of decision variables. Recently, Poonthalir and Nadarajan [163] intended to address the issues that are associated with the waiting time at the fueling station. Each fueling station was modeled as M/M/1 queue model, where the vehicle is allowed to wait in the queue.

In GVRP, alternative-fueling stations are copied multiple times and treated as special customers (dummy nodes) which are visited at most one time. Therefore, the GVRP was converted to be under the traditional VRP frameworks and it can be solved by employing efficient heuristics proposed for VRPs. Schneider et al. [184] extended GVRP with considering time window constraint and recharging stations. They developed TS with a variable neighborhood algorithm to solve the problem and got better results than density-based clustering and MCWS algorithms presented by Erdogan and Miller-Hooks [48]. Later, Schneider et al. [185] developed VRP with intermediate stops and solved the problem using an adaptive variable neighborhood search algorithm. Montoya et al. [140] developed a simple but effective two-phase heuristic to solve the GVRP. They also performed extensive experiments on 52 instances from the literature to test their heuristic.

Frank et al. [63] introduce an arc-duplicating formulation for a rich VRP with time windows, considering capacity restrictions including cargo and energy, customer time windows, and possible recharging stops. Koç and Karaoglan [94] adapt the arc-duplicating formulation of Frank et al. [63] to the GVRP. They used a SA based exact solution approach using branch and cut technique to solve the proposed problem. Yavuz [219] developed an iterated beam search algorithm for the GVRP based on arc duplication.

In the literature, there are some papers that have used set partitioning to solve GVRP types. Montoya et al. [140] proposed a multi-space sampling heuristic for GVRP. In particular, a randomized route-first cluster-second heuristic is firstly used for generating a set of feasible single-vehicle routes. Then, the feasible set is fed into a set partitioning formulation for determining a solution of the GVRP. The model focused on minimizing the length of the route and carbon dioxide-equivalents emission. They showed that it outperforms all other heuristics and meta-heuristics designed for the GVRP. Very recently, in [80], an extended fleet for including conventional, PHEVs, and EVs was proposed which was heuristically solved via a hybrid GA enhanced with local and LNS and set partitioning. Exact solution via branch-and-price-and-cut has also been developed.

Desaulniers et al. [44] tackled four problem variants with time windows, which are the combinations of single and multiple visits of a recharging station and partial and full charge allowances, while Andelmin and Bartolini [7] tackled the basic form of GVRP. A key difference between two proposed exact approaches for solving these problems is that Desaulniers et al. [44] used node duplication

in their pricing problem, whereas Andelmin and Bartolini used arc duplication in a way that includes multiple external stations visits back-to-back under the fixed-full refueling approach. They formulated a set partitioning problem where columns illustrated the feasible routes like simple circuits in multiple graphs in which each node is associated with a customer while an arc represents a non-dominated path between two customers that possibly uses AFSs. The set partitioning formulation is strengthened through several valid inequalities and it is then solved with a CG method. Their computational experiments showed that the proposed exact solution approach is suitable to optimally solve some large-scale instance sets in an average computational time of 3 h. As reported in [115], CG approaches are very effective but their convergence is very slow. Thus, they are generally suitable to solve large-sized instances in long computational times, but they are not suited to solve smaller-sized instances in very fast computational times. Fukasawa et al. [68] proposed a branch-and-cut-and-price algorithm for the energy minimizing vehicle routing problem (EMVRP) in which the energy consumption over each arc is defined as the product of the arc length and the weight of the vehicle. They developed two new MILP formulations for the problem: an arc-load formulation and a set partitioning formulation.

A path-based MILP formulation was done for GVRP by Pisacane et al. [159]. They generated all feasible routes and eliminated all dominated paths from the feasible set which is given as an input to a set partitioning formulation that gives solutions to the problem. Leggieri and Haouari [101] proposed a new MILP of GVRP for considering time and energy consumption constraints that is neither node- nor arc-duplicating. Their formulation reduced the use of various variables and constraints and included a set of preprocessing conditions which made the problem to be efficiently solvable using commercial solvers.

Another problem of GVRP dealing with the recharging or refueling of the vehicles, particularly the AFV, is charging infrastructure planning (e.g., [75]). Government agencies, nonprofit organizations, municipalities, and some private companies have started to convert their fleets of trucks to AFVs so as either to satisfy the energy policies or environmental regulations or to voluntarily reduce the environmental impact [48]. GVRP is mainly focused on alleviating the negative influences on the environment associated with conventional vehicles. One way of alleviating such impacts is to employ the AFVs. Compared with conventional vehicles, the AFVs use clean energy like electricity and hydrogen as power sources. However, the AFVs have some shortcomings such as limited energy storage (e.g., low battery capacity of electric vehicles) and access to AFSs. Consequently, the refueling issue becomes more important in solving the routing problem of the AFVs, which makes the route selection for the AFVs complicated. The problems with capacitated AFSs have been introduced also for the GVRP by Bruglieri et al. [27]. In [130], two MILPs have been proposed for routing a set of AFVs considering pickup and delivery operations in a semiconductor supply chain. In Soysal and Çimen [37], a green time-dependent capacitated vehicle routing problem (CVRP) has been addressed through dynamic programming.

The time-dependent GVRP literature has mainly focused on time-dependent travel times due to congestion on roads. The studies can be classified as static or

dynamic if the times are fixed for a given time interval, or change dynamically, respectively. Static mathematical models include deterministic [62, 83, 93, 162, 170, 192, 211] or stochastic features [37, 169] whereas [191] dealt with dynamic models. A periodic GVRP with time-dependent urban traffic and time window was studied by Mirmohammadi et al. [139]. They modeled the problem to minimize carbon emission, earliness, and lateness penalties and demonstrated their results for some generated test problem instances.

1.3.1 Transportation and Energy Consumption

Transportation, which is one of the most important parts of logistics, is the irreplaceable fundamental infrastructure for economic growth. However, it is also one of the hugest petroleum consumers and accounts for a large part of the overall pollutants [175]. Some green routing problems focus on energy consumption. Reducing fuel consumption may clash with the designated economic objectives. Exploring the relationship between environmental effect and transportation through route planning will be able to provide practical and valuable suggestions. It is also desirable that a decrease in petroleum-based fuel consumption can correspondingly reduce the greenhouse gas emission significantly [215]. Therefore, fuel consumption is an important index in the GVRP [126]. To include the fuel consumption in the routing model, the formulation of computing fuel consumption concerning the condition of a traveling vehicle is essential. According to the report by the US Department of Energy (2008), the weight of the load, travel speed as well as the traveled distance are influential parameters and the key factors that considerably affect the amount of the fuel consumption.

Some studies about the fuel consumption model in terms of transportation exist in the literature, which provides relevant references to the research on GVRP. Just to cite a few, a formulation of fuel consumption is provided in [215]. They proposed an ECR that extends CVRP with the aim of minimizing fuel consumption. In the paper, both the load and the distance traveled are considered as the factors which determine the fuel costs. Fuel consumption rate is taken as a load-dependent function, where the consumption rate is linearly associated with the vehicle's load. Tiwari and Chang [200] considered a GVRP model with the emissions minimizing objective and estimated the carbon dioxide emissions based on truckload and traveling distance. A block recombination approach is used to solve the proposed problem. In addition to the loading weight and the transportation distance that are considered in the above two papers, Kuo [126] added the transportation speed to the fuel consumption calculation model in time-dependent VRPs. Other VRP-related researches that aim at minimizing total fuel consumption are [8, 50, 135, 145, 176, 198]. Demir et al. [41–43], and Bektas and Laporte [17] applied a comprehensive mathematical model for estimating fuel consumption as an important parameter for CO_2 emission. Scott et al. [187] and Jovicic et al. [88] developed a computer application to import data from the road and measure directly fuel consumption. Also, Jabali et al. [83] and Pan et al. [150] proposed a new approach to calculate the fuel consumption which

is proportional to the amount of CO_2 emission. Some survey papers that focus on different fuel consumption formulations are [18, 43]. To decrease the amount of fuel consumption in a time-dependent problem, Kuo [126] applied a string-based SA on retail stores. Their formulating is based on the calculation done by Kuo [126]. Suzuki [194] investigated a multi-stop truck routing problem consisting of time-constrained transportation which aims to reduce fuel consumption. Xiao et al. [215] presented a time-dependent mathematical model that finds the best routes with the lowest fuel consumption and pollutant emission. To calculate fuel consumption, the authors employed a regression formulation which had an improvement of 24.61% in fuel consumption with better routing plans but longer transportation time and distance. Since this consumption model does not use the deceleration, recently Suzuki and Lan [196] extended it, by including negative values of acceleration to study the fuel consumption of trucks in congested areas. They ensured that the fuel use rate cannot be lower than that of engine idling, to avoid the unreal case of negative fuel burn during the deceleration phase.

Kara et al. [117] examined developing an EMVRP which focused on minimizing the weight of loads that leads to decreasing fuel consumption. To solve a model for minimizing fuel consumption in a routing problem with a time window, Li [102] developed a TS algorithm with random variable neighborhood decent procedure. Teng and Zhang [199] proposed a GVRP with a load factor that tries to minimize driving distance and total fuel cost and solved using SA. Jabir et al. [84] developed a hybrid ACO for capacitated multi-depot GVRP to decrease emission cost. Kazemian and Aref [120] solved a capacitated time-dependent mathematical model that aims to minimize both fuel consumption and fuel emission.

1.3.2 The Pollution-Routing Problem

The growing concerns about direct or indirect hazardous effects of transportation on humans and the whole ecosystem require re-planning of the road transport network and flow by explicitly considering Greenhouse Gas (GHG) and in particular CO_2 emissions [17]. The Pollution-Routing Problem (PRP) aims at choosing a vehicle dispatching scheme with less pollution, in particular, reduction of carbon emissions. Reducing CO_2 emissions is achievable by extending the traditional routing problem to take into account relevant environmental and social impacts [40, 69, 131, 137, 149, 167, 168, 179]. Although the traditional routing problem objective of minimizing the total distance will in itself contribute to a decrease in fuel consumption and pollutant emissions, this relationship needs to be directly calculated using more accurate formulations.

Pronello and André [165] suggested that reliable models for estimating the pollution generated by vehicle routes need to consider traveling time when the engine is cold. Sbihi and Eglese [179] considered a time-dependent routing problem in the context of traffic congestion. Since less pollution is produced when the

vehicles are at best speeds, directing them away from congestion tends to be more environmentally friendly, even though it leads to longer traveling distance. Also, Maden et al. [131] solved the same problem and reported about 7% reduction of CO_2 emissions based on a function conducted from the time-varying speeds. However, the objective function of their model remained the minimization of the total travel time instead of reducing emissions. Palmer [149] developed a routing model and calculated the amount of carbon dioxide as well as distance and traveling time. The paper examined how speed affects the reduction of CO_2 emissions in different congestion scenarios with time windows.

A fuel emission minimization model that concentrated on reducing emission for solid waste collection trucks was done by Apaydin and Gonullu [8]. An emission minimization problem was proposed by Figliozzi [58] that examined routing vehicles with less emission where emission minimization is taken as an additional objective. The results illustrated about 5% reduction of the CO_2 emissions emitted on the journey. Bauer et al. [15] explicitly concentrated on minimizing GHG emissions in a model of intermodal freight transport, showing the potential of intermodal freight transport for reducing greenhouse emissions. Fagerholt et al. [51] tried to reduce fuel consumption and fuel emissions by optimizing the speed in a shipping scenario. Given the fixed shipping routes and the time windows, the speed of each segment of a route is optimized to yield fuel savings.

Ubeda et al. [202] is the first study that incorporates minimizing GHG emissions in the model of routing problem with backhauls. They conducted a comprehensive objective function that measures both the distances and pollutant emissions. The results also revealed that backhauling seems more effective in controlling emissions. The authors suggest that backhauling can be initiated by companies to enhance energy consumption efficiency and reduce environmental impact. Faulin et al. [52] developed a CVRP with environmental criteria and considered a more complex environmental impact. Apart from the traditional economic cost's measurement and the environmental costs that are caused by polluting emissions, the environmental costs derived from noise, congestion, and wear and tear on infrastructure were also considered.

Following up Bektas and Laporte [17] and Demir et al. [41] proposed an extended ALNS for PRP to minimize fuel consumption, emission, and driver cost and enhance the computation efficiency for medium or large-scale PRP. The algorithm for solving PRP iterates between solving VRPTW and speed optimization. Initial routes are formed for VRPTW using fixed speed and are improved using a speed optimization algorithm. They further extended the work to solve PRP as a bi-objective optimization problem [42] that concentrated on arriving at a tradeoff between the conflicting objectives of fuel consumption and total driving time. The problem is modeled with four different a posteriori methods using an enhanced ALNS procedure where speed optimization is done at each iteration. They have used a comprehensive emission model to estimate fuel consumption.

A bi-objective GVRP using NSGA-II was formulated by Jemai et al. [85] with objectives to minimize distance and CO_2 emission. A green vehicle distribution model for the public transport network was developed as a nonlinear optimization

problem by Jovanovic et al. [87]. They designed an Adaptive Neuro-Fuzzy Inference System (ANFIS) model with environmental parameters, passenger costs, and their impact on green routing to solve the problem. Cirovic et al. [38] concentrated on routing light delivery vehicles using an adaptive neural network that focused on reducing air pollution, noise level, and logistics operating costs. They pondered routing limited environmentally friendly vehicles and unfriendly vehicles separately. The input parameters for the neural network were logistics operating costs and environmental parameters which were used to assess the performance of network links for calculating routes and then a MCWS algorithm was used. A Satisfactory-GVRP was developed by Afshar-Bakeshloo et al. [4] as an extension of PRP that takes into account the economic, environmental, and customer satisfaction as objectives.

Fukasawa et al. [69] developed two mixed-integer convex optimization models for the PRP. Dabia et al. [40] presented a branch-and-price algorithm for the PRP, and a new dominance criterion for the tailored labeling algorithm was developed to solve the pricing problem. Also, Majidi et al. [133] presented a nonlinear mixed-integer programming model for PRP. In another study, Franceschetti et al. [62] considered traffic congestion and proposed an enhanced ALNS algorithm for the time-dependent version of the PRP, which is an extension of [61]. Considering the demand and travel time uncertainty in green transport planning, Eshtehadi et al. [49] proposed several robust optimization techniques to solve PRP. Kargari Esfand Abad et al. [119] proposed a bi-objective model for pickup and delivery PRP to minimize the total system cost and fuel consumption to reduce the carbon emission. Van den Hove et al. [204] described the development of a cyclist routing procedure that minimizes personal exposure to black carbon. They indicated that the average exposure to black carbon and the exposure to local peak concentrations on a route are competing objectives, and proposed a parametrized cost function for the routing problem that allows for a gradual transition from routes that minimize average exposure to routes that minimize peak exposure.

All of the above studies are based on the assumption that there is a homogeneous fleet of vehicles. The PRP is one important variant of the GVRP that specifically necessitates a heterogeneous fleet of vehicles; firstly, because vehicles may produce a variety of emissions; and secondly, because logistic agencies may acquire heterogeneous vehicle types involving various categories of engines (fuel, electric, or hybrid), recent and old models, as well as vehicles from different brands [89, 95, 98, 127]. Pitera et al. [160] formulated an emission minimization routing decision for urban pickup system with a heterogeneous fleet. Their study demonstrated a significant improvement in emission reduction. Xiao and Konak [214] presented a time-dependent heterogeneous green vehicle and scheduling problem, with the objectives to minimize CO_2 emission and weighted tardiness. Travel schedules of vehicles were determined using traveled distance in different periods and solved using a dynamic programming approach. Yu et al. [224] propose an improved branch-and-price algorithm to precisely solve the heterogeneous fleet green vehicle routing problem with time windows. The latest studies about the

heterogeneous GVRP, where CO_2 emissions reduction are considered, can be found in the work of Yu et al. [225] and Li et al. [105].

The alternative fuel in routing problems is also studied under the category of the PRP. Although, the models in the PRP commonly consider a constant ECR and assume a linear relationship between energy consumption and the traveling distance of vehicles. Bektas and Laporte [17] proposed the PRP that considers both speed and load as main factors in emission and introduced a comprehensive mixed-integer nonlinear programming model. The objective function presented the minimization of the cost of carbon emissions along with the operational costs of drivers and fuel consumption. However, their model assumed a free-flow speed of at least 40 km/h, which was contrary to the real-world situation where congestion occurs. They adopted a fuel consumption function that is inspired by the comprehensive modal emission model (CMEM) developed by Barth et al. [13]. The main advantage of using the CMEM is twofold; first, the CMEM depends on the fuel type; second, biodiesel is viewed as an alternative fuel that can be used in conventional diesel engines, either on its own or mixed with diesel [206]. Besides, the properties and suitability of biodiesel do not require any adjustments in the engine of a specific vehicle due to the use of biodiesel instead of (or blended with) petroleum-based diesel. Thus, the new diesel-powered vehicles are designed to operate on biodiesel without modification [16]. For additional information on biodiesel fuels, the readers are referred to [132, 217].

1.4 Hybrid Vehicle Routing Problem

When a vehicle equipped with artificial fuel needs to take a long-distance tour, it has to halt for refueling in an alternate fueling station. When AFS is limited, a vehicle has to take up a route forcefully to refuel in the limited number of AFS which can increase the route cost. This problem can be tackled if the vehicle can switch between battery and fuel depending on the requirements, called the Hybrid Electric Vehicle (HEV), as proposed by Mancini [134]. In [134], a MILP formulation of the HVRP is carried out where the decision process is related to deciding both when to recharge and when to switch from an ICEV to an EV. The author proposes a large neighborhood search based metaheuristic and she also uses the GVRP benchmark instances for the experimental campaign.

Although, a HEV is preferable because a pure electric vehicle is limited to the availability of recharging stations and distance coverage, the availability of a routing model for HEVs is even less or quite absent. Therefore, this section is motivated to review the HVRP which is classified into the GVRP group by Lin et al. [112] as it also takes into account the environmental impact. HVRP aims to minimize the total travel cost incurred by utilizing hybrid vehicles. The practicality of HVRP is supported by the growing number of HEVs in the automobile market. For example, Juan et al. [89] introduced a VRP with multiple driving ranges, which was motivated by the practical challenge driven by the increasing number of EVs with different

degrees of autonomy in corporate fleets. They described an integer programming formulation and a multi-round heuristic algorithm that iteratively constructed a solution for the problem. Using a set of benchmarks adapted from the literature, the algorithm was then employed to analyze how distance-based costs are increased when considering "greener" fleet configurations.

Lin et al. [234] combined the practical application of HEVs in the logistics industry and studied the HVRP considering battery capacity constraints, gasoline constraints, and mode selection in HEVs based on the background of green logistics and the characteristics of HEVs, which minimized total energy consumption cost. The mode selection system can adjust the driving mode of the HEV according to different road conditions to obtain the optimal use of fuel. This problem was formulated as a mixed-integer linear programming model. To solve the problem, an improved PSO algorithm is developed in which a labeling procedure is involved.

While limited work is devoted to studying the HVRP, several versions of the classical VRP share features of the HVRP. For example, the GVRP presented by Erdogan and Miller-Hooks [48], Schneider et al. [184], and Hiermann et al. [79] entails scheduling efficient routes for EVs that are required to stop at charging stations distributed in the network to recharge their batteries. Failing to schedule proper stops for battery recharging precludes the vehicles from completing their scheduled tour and/or returning to their depots. As such, scheduling stops for battery recharging is a hard constraint for the vehicles in the GVRP. A similar constraint should be considered for the vehicles in the HVRP as their batteries need to be recharged to complete their tours. Besides, the HVRP is similar to the CVRP, where each vehicle has a limited carrying capacity [19, 35, 67, 215]. All studies on HVRP are shown in Table 1.6.

Several sources of complexity characterize the HVRP. First, even for small size problems, the HVDRP involves a large number of decision variables including vehicle resources/capabilities, locations of vehicle dispatching and collection, and routing decisions for the vehicles. The problem can be generally viewed as an extension of the classic VRP which is known to be an NP-hard problem [72]. Thus, the execution time required to obtain an exact optimal solution grows exponentially

Table 1.6 The studies on the HVRP and its variants

Year	Number of papers	The list of the papers
2007	2	[33, 186]
2008	1	[74]
2009	2	[20, 190]
2010	1	[6]
2013	1	[143]
2014	2	[77, 89]
2015	1	[109]
2016	1	[146, 193]
2017	2	[134, 223]
2019	4	[106, 107, 118, 234]

as the problem size increases. Second, most decision variables involved in this problem are highly interdependent and cannot be optimized separately.

A hybrid vehicle routing model was set up by Liu et al. [109] to solve the hot strip mill scheduling problem which was a combination of variable fleet VRP, prize collection vehicle routing, and CVRP. The constraints in hot mill processes were analyzed in three aspects: process scheduling, product means, and energy consumption. They proposed a new algorithm combining PSO and SA to solve a multi-objective optimization problem. The PSO solution performance was an imbalance phenomenon between the convergence rate and the convergence precision, due to the inherent mathematical operators of the PSO algorithm. Recently, Li et al. [107] minimized total travel costs in consideration of both electric charging stations and fuel stations. To solve the problem, they proposed a novel hybrid metaheuristic approach which was a hybridization of the memetic algorithm with sequential variable neighborhood descent.

There are two basic configurations of HEV: a series hybrid and a parallel hybrid. In a series hybrid, the power transmission comes from both combustion engines, and the drive system is from electricity. In a parallel hybrid, the power transmission comes from both combustion engines, and the electrical drive system is mechanical. These systems can be applied in combination or independently. Another hybrid configuration is a complex configuration, which involves a setting that cannot be classified into those two types mentioned. The complex hybrid seems to be similar to the series–parallel hybrid because the generator and electric motor are both electrical and mechanical. However, the key differences are the bidirectional power flow of the electric motor in the complex hybrid and the unidirectional power flow of the generator in the series–parallel hybrid [33, 77] classified HEV, based on the hybridization factor, into Micro-HEVs, Mild-HEVs, Power-assisted HEVs, and Plug-In HEVs (PHEVs).

A PHEV is the next generation of hybrid electric vehicles, offering significant advantages over the cleanest and most efficient of today's hybrid vehicles [6]. PHEV's sources of energy for propulsion consist of electricity from the grid, energy recovery while braking (which is stored in the battery), and chemical energy from a hydrocarbon-based fuel stored in the fuel tank [190]. Bradley and Frank [20] explained that a PHEV can work in several modes, such as charge-depleting mode, charge-sustaining mode, EV mode, and engine-only mode. Silva et al. [190] classified PHEV into a range extender PHEV and a blended PHEV. A range extender PHEV runs primarily on electricity discharged from the battery, and after the SOC reaches a minimum, it operates in a charge-sustaining mode. A blended PHEV is similar to the range extender PHEV, but in the charge-depleting mode, it may turn on the internal combustion engine whenever needed.

Yu et al. [223] focused on vehicles that use a hybrid power source, known as the PHEV and generate a mathematical model to minimize the total cost of travel by driving PHEV. They considered a time constraint and two types of energy resources: electricity and fuel. To solve the problem, they developed SA with a restart strategy.

The proposed SA algorithm was first verified with benchmark data of the CVRP, with the result showing that it performs well and confirms its efficiency in solving CVRP.

The routing algorithms for PHEVs should also account for the significant energy efficiency differences of different operating modes and recommend the predominant mode of operation for each road segment during route planning. The energy efficiency differences are also a function of the vehicle (e.g., payload) and road segment features (e.g., speed limits, terrain geometry). Given that internal combustion engines tend to be most efficient when operating at steady highway speeds of 45–65 mph (miles per hour), the electric mode is relatively efficient on city roads with lower speed limits [161].

The strategy to control the energy among these multiple energy sources of a hybrid vehicle is termed "powertrain energy management." An overview of this area is provided by Sciarretta and Guzzella [186]. Currently, powertrain energy management control algorithms for PHEVs are mostly "static" and try to utilize the charge within the battery as soon as possible, without explicit consideration for real opportunities that might be present within the route to best utilize the battery charge. This is also attributable to the fact that there exist no route planning algorithms for PHEVs, and automotive original equipment manufacturers have yet to develop "dynamic" energy management control algorithms that account for the complete route plan in controlling the vehicle powertrain operating modes. Several researchers investigated the powertrain management for hybrids and PHEVs from an energy management control perspective (e.g., [74, 143]). Nejad et al. [146] addressed the energy-efficient routing of PHEVs during route planning, which is a higher-level energy management process for PHEVs and designed exact and approximation routing algorithms that consider general graphs.

The existing shortest-path based algorithms cannot be applied to solving problems for PHEVs, backup batteries with combustion engine, because of the several new challenges: (1) an optimal route may contain circles caused by detour for recharging; (2) emissions of PHEVs not only depend on the driving distance, but also depend on the terrain and the SOC of batteries; (3) batteries can harvest energy by regenerative braking, which makes some road segments have negative energy consumption. To address these challenges, Sun and Zhou [193] proposed a green navigation algorithm (GNA) that found the optimal strategies: where to go and where to recharge. GNA discretized the SOC, then made the PHEV routing problem to satisfy the principle of optimality. Finally, GNA adopted dynamic programming to solve the problem.

1.5 Conclusions

It is evident that the managing of fuel consumption and the use of sustainable energy sources in road logistics and transportation is more necessary than ever, and constitutes a critical factor for the evolution of an economically and environmentally

stable world. This chapter has reviewed some of the existing literature related to three major and applicable streams of the VRPs, namely, EVRP, GVRP, and HVRP. The main objective of this chapter was to organize the available literature so as to provide a reference point for future research on new aspects of routing problems. A wide range of these three variants and their complicate planning efforts have been studied. We have presented a comprehensive and up-to-date review of the existing studies by paying special attention to environmental issues. The following conclusions and future research perspectives are identified:

1.5.1 Future Research Directions in EVRP

Although there is a large literature related to the EVRP scheme, the EV network modeling study is limited and evolving. On this basis, the following research directions are proposed:

We believe that promising research avenues lie in the integration of additional features in the EVRP. For example, Jie et al. [86] did not consider all realistic situations in the problem. Future work can extend their model by considering the battery recharging time and customer service time window. The capacity limitation of satellites and BSSs can be incorporated also into the manuscript. Another interesting future work could be the development of the rewards-driven mechanism [70] or maintenance operation [46] in the field of EVRP, which can bring greater benefits to logistics companies. One natural extension could be to incorporate in-route recharging into the problem to render it applicable in mid- and long-haul contexts.

Handling the variation of energy consumption depending on different external factors is one of the most important challenges. Future studies may consider more practical factors related to the ECR function, such as the air conditioner status and environmental temperature. The development of more efficient solution approaches would also be welcome to solve large-sized instances with multiple vehicles. The development of new optimization and hybrid optimization-simulation methods to efficiently cope with the challenges of the incorporation of EVs in the road distribution activities, including dynamic scenarios especially with other types of uncertainty sets would be high value for promoting the desirable shift towards more sustainable energy sources in the logistics and transportation area.

1.5.2 Future Research Directions in GVRP

Based on the review on GVRP presented above, we draw the following conclusions about the trends of GVRP, through the analysis of how the GVRP can interact with the traditional VRP variants. There exist extensive VRP variants that can be combined with the GVRP model and make it comprehensive and closer and

more applicable to real-world problems. As the fuel consumption model is closely related to the condition of the vehicle, the flexibility offered by using different types of vehicles may result in more reduction of fuel consumption. The behavior of heterogeneous vehicles under varying speed environment is not still explored in the existing literature. Additional constraints like road conditions, vehicle load can be included with varying speed environment in fuel-efficient GVRP and their impact on route cost and fuel consumption can be studied. Also, the researches can be further extended to study the impact on route cost and fuel consumption in routing problems where speed variation can occur with partial and fully charged batteries. The model can also be used to study the impact of speed and fuel consumption with hybrid vehicles, when there is a switch over between battery and fuel which affects the speed.

Due to its significant practical relevance, the modeling of capacity restrictions at the recharging stations is another aspect that has gained increasing interest in the recent literature (see, e.g., [203, 228]). For instance, Upchurch et al. [203] developed the capacitated flow refueling location model that additionally integrates limited capacities for stations to limit the number of vehicles refueled at the same station. The approach of Chung and Kwon [36] considers the fact that the installation of a network of stations is a long-term process that usually requires several periods to be completed. Hence, the model restricts the number of newly established stations per period while pursuing maximization of possible traffic over all periods.

Since long waiting times for recharging would cause considerable customer inconvenience and, as a direct consequence, may even endanger the market adoption of electric cars, its modeling is essential. However, it is worth mentioning that this extension substantially changes the basic problem structure. Note that in the original refueling station location problem model, the coverage of demand depends only on the set of opened stations, but is not influenced by other demands. In the capacitated case, however, demands to be covered may compete for opened stations since their usage on necessary paths is limited. However, the development of algorithms that are based on sophisticated decompositions may, also, in this case, be a promising field of future research.

In the recharging problems, some restrictions in the real world have not yet been accommodated in the routing model. For example, the availability and the fuel capacity of the recharging stations will cause some change to the optimal routes. The stochastic service time of the recharging stations is also worth attention as it influences the time traveled and the arrival time at each point. An immediate extension of the GVRP is to model with techniques like queuing models that seem suitable for tackling the service time problem in this context.

Similar to the travel times, the service times of the customers may also vary depending on the time of the day according to several factors such as ease of access from the road and availability of parking spaces. We believe that the service time, especially in the stochastic form, is not negligible in future research and the real-time transportation information can lay a solid foundation for continued research into the PRP by providing dynamic real-world data. With the support of real-time information about traffic conditions, vehicles can be directed to other roads which

are less congested areas. This implies a more environmental-friendly case because less emission is generated when vehicles are traveling at the best speeds to meet service time exactly.

1.5.3 Future Research Directions in HVRP

Utilizing HVRP can help companies or organizations reduce their total travel cost by determining the optimal route that serves more consumers in less dependence on fossil fuels. Future developments in this field could address the definition of extensions of this problem in which traffic limited zones, in which it is forbidden to use traditional fuel propulsion are considered. Other developments could address the possibility of partial battery recharging and the usage of the so-called regenerative braking which allows EVs to recover a percentage of their battery charge while traveling on downhill roads. Moreover, multiple recharging technologies, with different recharging times, costs, and availability could be considered, as in the work by Felipe et al. [53] on the pure electric propulsion case.

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